

Performance Testing

Date	7 July 2026
Team ID	SWTID-2026-3702
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (API testing) + Manual browser testing
Type of Testing	Functional Testing, Unit Testing, Integration Testing
Target Module / API	Flask routes: / (home), /Prediction (form), /predict (POST)
Test Environment	Local development — Windows 10, Python 3.10, localhost:5000
Test Date	June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	Input Values	Expected Outcome
1	Predict Very High HDI	Country=Norway(128), Life Exp=82, Schooling=13, GNI=67614, Internet=97	Very High HDI + score ~0.90+
2	Predict Medium HDI	Country=India(76), Life Exp=69, Schooling=6.5, GNI=6590, Internet=45	Medium HDI + score ~0.60
3	Predict Low HDI	Country=Afghanistan(0), Life Exp=54, Schooling=3.5, GNI=1900, Internet=15	Low HDI + score ~0.35-0.45
4	Single value test	[13, 72.0, 5.2, 3341.0, 14.4]	Score: ~0.59 (Medium HDI)
5	Missing/empty field	Empty GNI field submitted	Flask error handling / validation
6	Home page load	/	home.html renders correctly with navbar
7	Prediction page load	/Prediction	indexnew.html renders with form fields
8	Navigation links	Click Home/Predict links	Correct page rendered

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	< 0.5 seconds	Pass	Flask model loaded at startup
2	Response Time (Max)	< 5 seconds	< 1 second	Pass	Pickle load is very fast (732 bytes)
3	Model R-squared Score	> 90%	93.74%	Pass	Linear Regression performs well
4	Error Rate	< 1%	0% on valid inputs	Pass	All valid inputs processed
5	CPU Utilization	< 80%	< 5% (local)	Pass	Lightweight Flask application
6	Memory Utilization	< 80%	< 100 MB	Pass	Minimal memory footprint
7	Home Page Load	< 1 second	< 0.3 seconds	Pass	Static HTML rendering
8	Prediction Accuracy	R-squared > 0.90	0.9374	Pass	Model predicts HDI within ± 0.05

Step 4: Observations & Analysis

Observation	Finding
Model Performance	Linear Regression achieves R-squared = 93.74%, explaining 93.74% of variance in HDI scores. Predictions are within ± 0.05 of actual values for most test cases.
Application Speed	Flask application responds in under 0.5 seconds for all prediction requests. Pickle model loading at startup eliminates per-request retraining overhead.
Category Classification	HDI category mapping (Low/Medium/High/Very High) correctly classifies 19/20 test countries based on predicted scores.
Edge Cases	Out-of-range input values (e.g. negative life expectancy) return the 'values do not match range' message, demonstrating basic error handling.
Scalability Note	Current single-model architecture handles individual predictions well. For batch processing or concurrent users, a task queue (Celery) should be added.